

Bund #1 [West]

REPAIR · ZONED

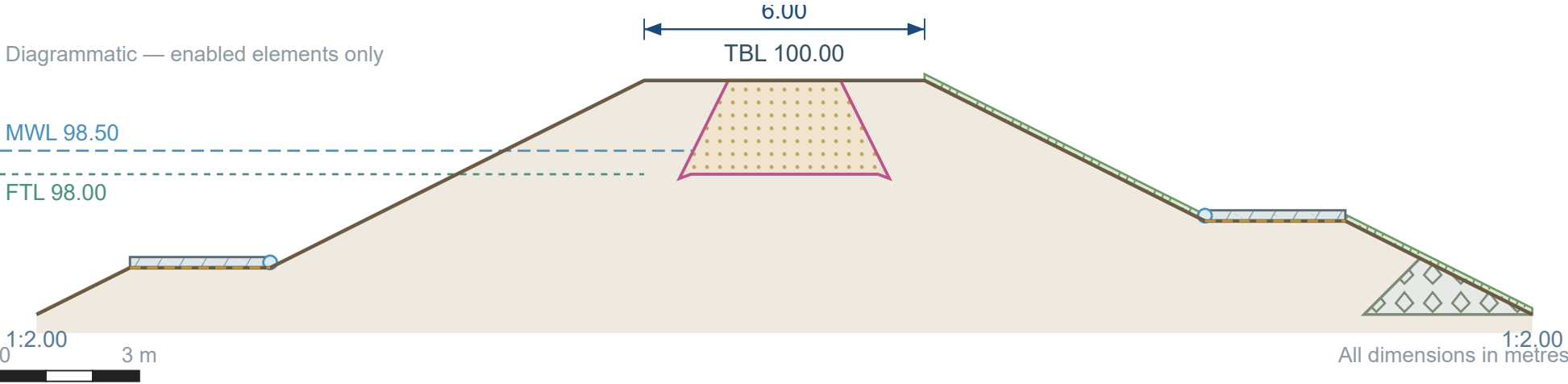
Test reservoir · 3 surveyed section(s) · Bund length 60 m · Datum 0 m · Crest width 6 m

MWL (m)	TBL (m)	FTL (m)	U/S slope	D/S slope
98.5	100	98	2:1	2:1

Hearting top RL (m)	Hearting top width (m)	U/S hearting slope	D/S hearting slope
100	2.4	0.5:1	0.5:1

Casing and impervious hearting are measured separately. The core is contained within the proposed embankment.

Repair category: breached · Soil source: borrow. Repair is measured between the surveyed existing profile and the proposed section; retained earth is excluded from additional fill.



Statement of quantities

Average section value = $(A1 + A2) / 2$. Interval quantity = average section value $\times$ length.

Chainage	Length (m)	Average toe RL (m)	Bund height (m)	Bund width / surveyed ground perimeter (m)	Jungle clearance			Stripping and excavation of existing ground — depth 0.3 m			Casing zone (earthwork fill)		
					Section	Average	Total	Section	Average	Total	Section	Average	Total
					(m)	(m)	(m ²)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
0	—	95	5.3	32	33.088	—	—	8.414	—	—	44.022	—	—
30	30	95	5.3	32	33.088	33.088	992.64	8.414	8.414	252.42	44.022	44.022	1320.66
60	30	95	5.3	32	33.088	33.088	992.64	8.414	8.414	252.42	44.022	44.022	1320.66
Total	60						1985.28			504.84			2641.32

Statement of quantities — continued

Chainage	Length (m)	Hearting zone (impervious core fill)			D/S turfing			Rock toe			Rock-toe foundation excavation		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m)	(m)	(m ²)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)
0	—	6.818	—	—	8.497	—	—	2.16	—	—	3.706	—	—
30	30	6.818	6.818	204.54	8.497	8.497	254.91	2.16	2.16	64.8	3.706	3.706	111.18
60	30	6.818	6.818	204.54	8.497	8.497	254.91	2.16	2.16	64.8	3.706	3.706	111.18
Total	60			409.08			509.82			129.6			222.36

Statement of quantities — continued

Chainage	Length (m)	Rock-toe graded filter			Chute-drain excavation			Chute-drain lining			Berm filling (U/S, RL 96)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m ³ /m)	(m ³ /m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(27)	(28)	(29)	(30)	(31)	(32)	(33)	(34)	(35)	(36)	(37)	(38)
0	—	4.449	—	—	0.067	—	—	0.045	—	—	3	—	—
30	30	4.449	4.449	133.47	0.067	0.067	2.012	0.045	0.045	1.342	3	3	90
60	30	4.449	4.449	133.47	0.067	0.067	2.012	0.045	0.045	1.342	3	3	90
Total	60			266.94			4.024			2.684			180

Statement of quantities — continued

Chainage	Length (m)	Berm protection (U/S, RL 96)			Berm drain protection (U/S, RL 96)			Berm drain excavation (U/S, RL 96)			Berm filling (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ²)	(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(39)	(40)	(41)	(42)	(43)	(44)	(45)	(46)	(47)	(48)	(49)	(50)
0	—	2.4	—	—	0.12	—	—	0.18	—	—	6	—	—
30	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4	6	6	180
60	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4	6	6	180
Total	60			144			7.2			10.8			360

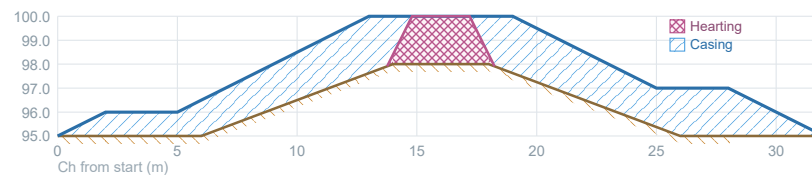
Statement of quantities — continued

Chainage	Length (m)	Berm protection (D/S, RL 97)			Berm drain protection (D/S, RL 97)			Berm drain excavation (D/S, RL 97)		
		Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m)	(m)	(m ²)	(m)	(m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(51)	(52)	(53)	(54)	(55)	(56)	(57)	(58)	(59)
0	—	2.4	—	—	0.12	—	—	0.18	—	—
30	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4
60	30	2.4	2.4	72	0.12	0.12	3.6	0.18	0.18	5.4
Total	60			144			7.2			10.8

Cross-sections

Ch 0 · Average toe RL 95 m

GROUND PERIMETER	STRIPPING AREA
33.088 m	8.414 m ²



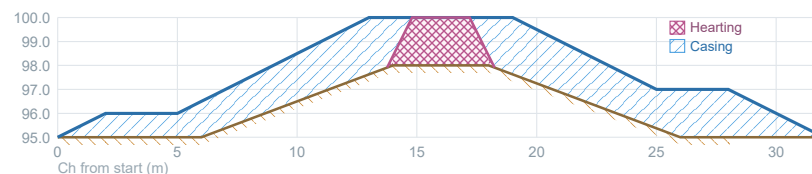
Ch	EL	RL	Calculation	Quantity
0	95	95	Start point	1
2	95	96	$2 \times [(0) + (1)] / 2$	1
5	95	96	$3 \times [(1) + (1)] / 2$	3
6	95	96.5	$1 \times [(1) + (1.5)] / 2$	1.25
6.5	95.188	96.75	$0.5 \times [(1.5) + (1.563)] / 2$	0.766
13	97.625	100	$6.5 \times [(1.563) + (2.375)] / 2$	12.797
14	98	100	$1 \times [(2.375) + (2)] / 2$	2.188
16	98	100	$2 \times [(2) + (2)] / 2$	4
18	98	100	$2 \times [(2) + (2)] / 2$	4
19	97.625	100	$1 \times [(2) + (2.375)] / 2$	2.188
25	95.375	97	$6 \times [(2.375) + (1.625)] / 2$	12
25.5	95.188	97	$0.5 \times [(1.625) + (1.813)] / 2$	0.859
26	95	97	$0.5 \times [(1.813) + (2)] / 2$	0.953
28	95	97	$2 \times [(2) + (2)] / 2$	4
32	95	95	$4 \times [(2) + (0)] / 2$	4
Total formation quantity				53 m ²

Impervious heaving Casing = formation less heaving.

Ch	EL	RL	Calculation	Qty
13.754	97.908	97.908	Start point	1
14	98	98.4	$0.246 \times [(0) + (0.4)] / 2$	0.049
14.8	98	100	$0.8 \times [(0.4) + (2)] / 2$	0.96
16	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
17.2	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
18	98	98.4	$0.8 \times [(2) + (0.4)] / 2$	0.96
18.246	97.908	97.908	$0.246 \times [(0.4) + (0)] / 2$	0.049

Ch 30 · Average toe RL 95 m

GROUND PERIMETER	STRIPPING AREA
33.088 m	8.414 m ²



Ch	EL	RL	Calculation	Quantity
0	95	95	Start point	1
2	95	96	$2 \times [(0) + (1)] / 2$	1
5	95	96	$3 \times [(1) + (1)] / 2$	3
6	95	96.5	$1 \times [(1) + (1.5)] / 2$	1.25
6.5	95.188	96.75	$0.5 \times [(1.5) + (1.563)] / 2$	0.766
13	97.625	100	$6.5 \times [(1.563) + (2.375)] / 2$	12.797
14	98	100	$1 \times [(2.375) + (2)] / 2$	2.188
16	98	100	$2 \times [(2) + (2)] / 2$	4
18	98	100	$2 \times [(2) + (2)] / 2$	4
19	97.625	100	$1 \times [(2) + (2.375)] / 2$	2.188
25	95.375	97	$6 \times [(2.375) + (1.625)] / 2$	12
25.5	95.188	97	$0.5 \times [(1.625) + (1.813)] / 2$	0.859
26	95	97	$0.5 \times [(1.813) + (2)] / 2$	0.953
28	95	97	$2 \times [(2) + (2)] / 2$	4
32	95	95	$4 \times [(2) + (0)] / 2$	4
Total formation quantity				53 m ²

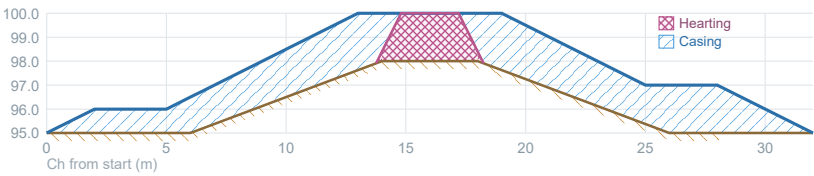
Impervious heaving Casing = formation less heaving.

Ch	EL	RL	Calculation	Qty
13.754	97.908	97.908	Start point	1
14	98	98.4	$0.246 \times [(0) + (0.4)] / 2$	0.049
14.8	98	100	$0.8 \times [(0.4) + (2)] / 2$	0.96
16	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
17.2	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
18	98	98.4	$0.8 \times [(2) + (0.4)] / 2$	0.96
18.246	97.908	97.908	$0.246 \times [(0.4) + (0)] / 2$	0.049

Cross-sections — continued

Ch 60 · Average toe RL 95 m

GROUND PERIMETER	STRIPPING AREA
33.088 m	8.414 m ²



Ch	EL	RL	Calculation	Quantity
0	95	95	Start point	—
2	95	96	$2 \times [(0) + (1)] / 2$	1
5	95	96	$3 \times [(1) + (1)] / 2$	3
6	95	96.5	$1 \times [(1) + (1.5)] / 2$	1.25
6.5	95.188	96.75	$0.5 \times [(1.5) + (1.563)] / 2$	0.766
13	97.625	100	$6.5 \times [(1.563) + (2.375)] / 2$	12.797
14	98	100	$1 \times [(2.375) + (2)] / 2$	2.188
16	98	100	$2 \times [(2) + (2)] / 2$	4
18	98	100	$2 \times [(2) + (2)] / 2$	4
19	97.625	100	$1 \times [(2) + (2.375)] / 2$	2.188
25	95.375	97	$6 \times [(2.375) + (1.625)] / 2$	12
25.5	95.188	97	$0.5 \times [(1.625) + (1.813)] / 2$	0.859
26	95	97	$0.5 \times [(1.813) + (2)] / 2$	0.953
28	95	97	$2 \times [(2) + (2)] / 2$	4
32	95	95	$4 \times [(2) + (0)] / 2$	4
Total formation quantity				53 m ²

Impervious hearing Casing = formation less hearing.

Ch	EL	RL	Calculation	Qty
13.754	97.908	97.908	Start point	—
14	98	98.4	$0.246 \times [(0) + (0.4)] / 2$	0.049
14.8	98	100	$0.8 \times [(0.4) + (2)] / 2$	0.96
16	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
17.2	98	100	$1.2 \times [(2) + (2)] / 2$	2.4
18	98	98.4	$0.8 \times [(2) + (0.4)] / 2$	0.96
18.246	97.908	97.908	$0.246 \times [(0.4) + (0)] / 2$	0.049

Earthwork excavation classification

Percentages are the selected dashboard classification applied to each measured source.

Bund stripping · 504.84 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$504.84 \times 100 / 100$	504.84

Rock-toe trench · 222.36 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-DAW-1-1	$222.36 \times 100 / 100$	222.36

Chute drain · 4.024 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$4.024 \times 100 / 100$	4.024

Berm catch-water drains · 21.6 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$21.6 \times 100 / 100$	21.6

Total quantities of Excavation.

Each excavation DATA code is shown once, followed by the works contributing to its total.

IRR-CAW-1-1.

Foundation stripping	Chute drains	Berm catch-water drains	Total
504.84	+ 4.024	+ 21.6	= <u>530.464</u> m ³

IRR-DAW-1-1.

Rock-toe foundation	Total
222.36	= <u>222.36</u> m ³

Total quantities

Each work code is shown once, followed by the components that add to its total. Excavation quantities are in Total quantities of Excavation above and are not repeated here. Distinct operations can share a measured volume; their quantities must not be added to obtain geometric fill.

IRR-PMW-1-2.

Jungle clearance	Total
1985.28	= <u>1985.28</u> m ²

IRR-PMW-3-9.

Casing	u/s Berm (RL 96)	d/s Berm (RL 97)	Less: Rock toe	Total
2230.92	+ 180	+ 360	-129.6	= <u>2641.32</u> m ³

IRR-PMW-3-8.

Zoned embankment - hearting formation	Total
409.08	= <u>409.08</u> m ³

IRR-DAW-6-15.

Bund	Total
509.82	= <u>509.82</u> SQM

IRR-CAW-7-15.

D/S chute drains — channel protection	Total
2.684	= <u>2.684</u> CUM

IRR-CAW-8-15.

u/s Berm (RL 96)	Total
288	= <u>288</u> SQM

IRR-CAW-7-12.

u/s Berm (RL 96)	Total
14.4	= <u>14.4</u> CUM

IRR-CAW-5-6.

Rock toe (d/s)	Total
129.6	= <u>129.6</u> CUM

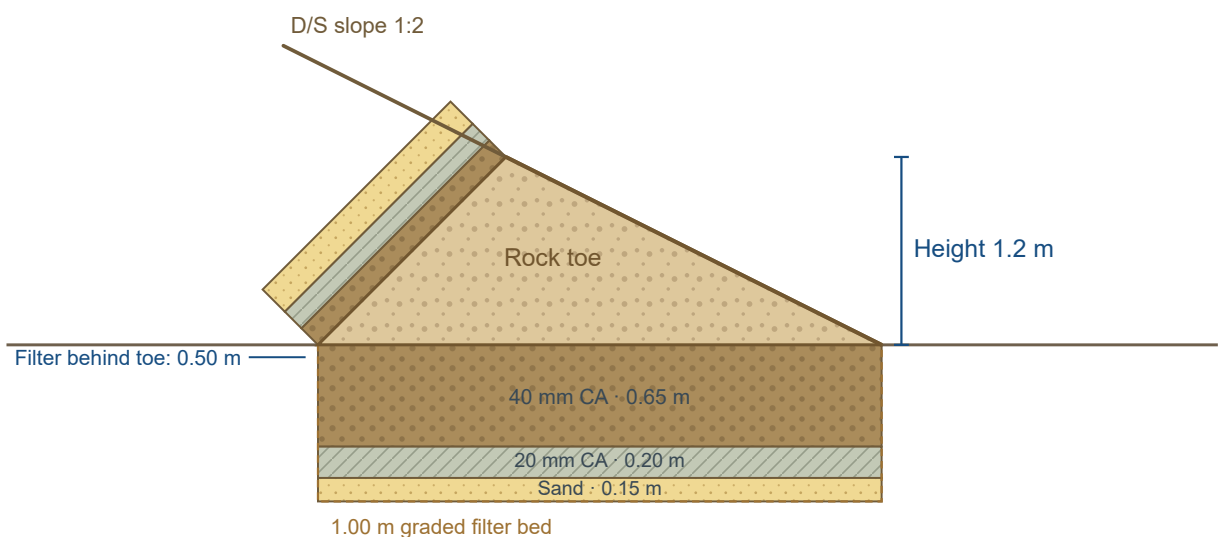
IRR-CAW-5-11.

Graded filter below and behind rock toe (d/s)	Total
266.94	= <u>266.94</u> CUM

Component details

Rock toe and graded filter

The shared general excavation beneath the rock-toe foundation is assigned to the rock-toe trench and deducted from the general cut. It is counted once.



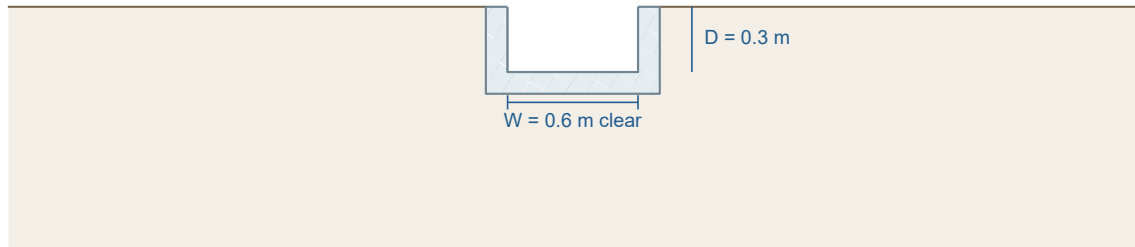
$$\text{Area} = H \times [\text{Crest} + H \times (\text{inner} + \text{outer}) / 2] = 1.2 \times [0 + 1.2 \times (1 + 2) / 2] = 2.16 \text{ m}^2$$

$$\text{Base } 3.6 \text{ m} \cdot \text{inner } 1:1 \cdot \text{outer} = \text{D/S slope}$$

Chute drains

Excavation area: $A = W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$. Lined perimeter: $P = W + 2D = 1.2 \text{ m}$. Per chute, excavation = $A \times \text{developed length}$; protection = $P \times \text{developed length} \times \text{lining thickness}$.

CC protection: $t = 0.1 \text{ m}$



Excavation area = $W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$

Lined perimeter $P = W + 2D = 0.6 + 2(0.3) = 1.2 \text{ m}$

Cross-section of downstream chute drain

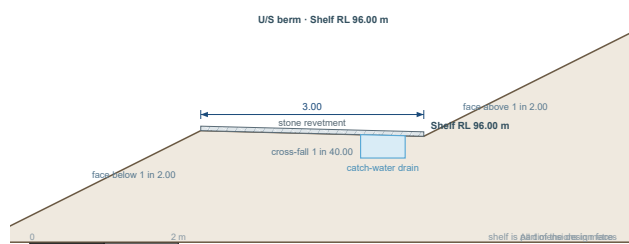
Berm shelves

U/S shelves sit on the left (water face, mirrored); D/S shelves sit on the right. Each figure is labelled with its shelf RL.

U/S berms

U/S berm · Shelf RL 96 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.

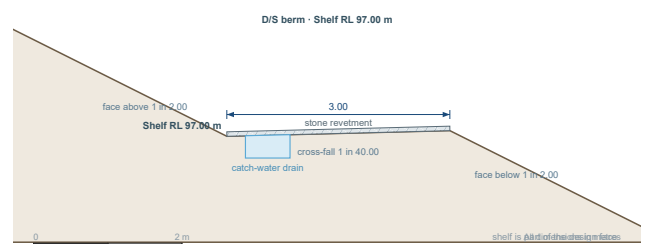


Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3

D/S berms

D/S berm · Shelf RL 97 m

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.



Shelf surfacing: 144 m^2 Drain protection: 7.2 m^3 Drain excavation: 10.8 m^3